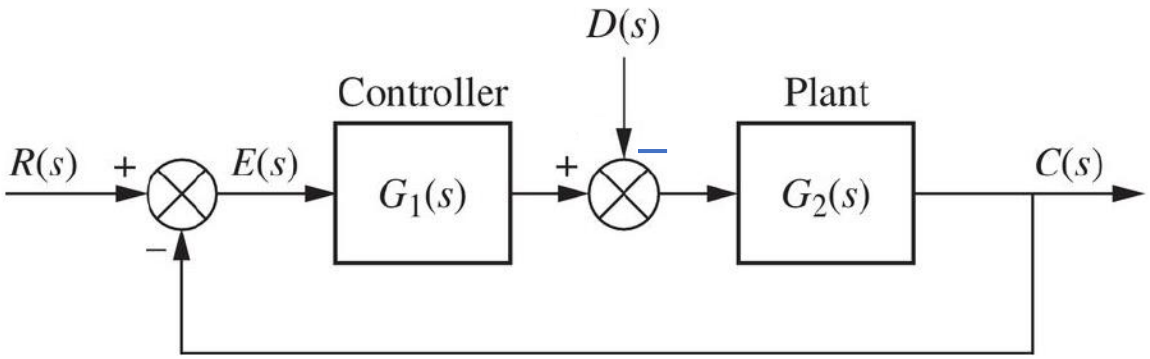
Steady State Error Summary
Main Input Error – Disturbance Error

Without loss of generality, for all purposes, we will refer to the following block diagram:



The total error $E(s)$ is given by

$$E(s) = \frac{1}{1 + G(s)} R(s) + \frac{G_2(s)}{1 + G(s)} D(s)$$

where $G(s) = G_1(s)G_2(s)$ is the open loop transfer function. The quantity

$$\frac{1}{1 + G(s)} R(s)$$

is the error due to the main input $R(s)$ or reference, and the quantity

$$\frac{G_2(s)}{1 + G(s)} D(s)$$

is the error due to the disturbance $D(s)$.

We can separate also the two steady state errors: the one caused by the reference and the one caused by the disturbance. The sum of these two errors produce the total error.

In previous class we found the following table related to the steady state error caused by the main input:

Type	Step Input $r(t) = r$	Ramp Input $r(t) = rt$	Parabola Input $r(t) = \frac{r}{2}t^2$
$G(s)$ is type 0	$\frac{r}{1+K_p}$	∞	∞
$G(s)$ is type 1	0	$\frac{r}{K_v}$	∞
$G(s)$ is type 2	0	0	$\frac{r}{K_a}$

Where

- Position Error Constant, $K_p = \lim_{s\to 0} G(s)$
- Velocity Error Constant, $K_v = \lim_{s\to 0} sG(s)$
- Aceleración Error Constant, $K_a = \lim_{s\to 0} s^2G(s)$

Also we have determined that the steady state error due to the disturbance can be summarized with the following table:

Disturbance	$G_1(s)$ Type 0	$G_1(s)$ Type 1	$G_1(s)$ Type 2
Step $D(s) = {}^{d}/_{s}$	$\frac{K_{p2}d}{1+K_{p1}K_{p2}}$ if $G(s)$ is type zero, or $\frac{d}{K_{p1}}$ if G(s) is type 1 or higher.	0	0
Ramp $D(s) = {}^{d}/_{s^2}$	∞	$\frac{d}{K_{v1}}$	0
Parabola $D(s) = {}^{d}/_{s^3}$	∞	∞	$\frac{d}{K_{a1}}$

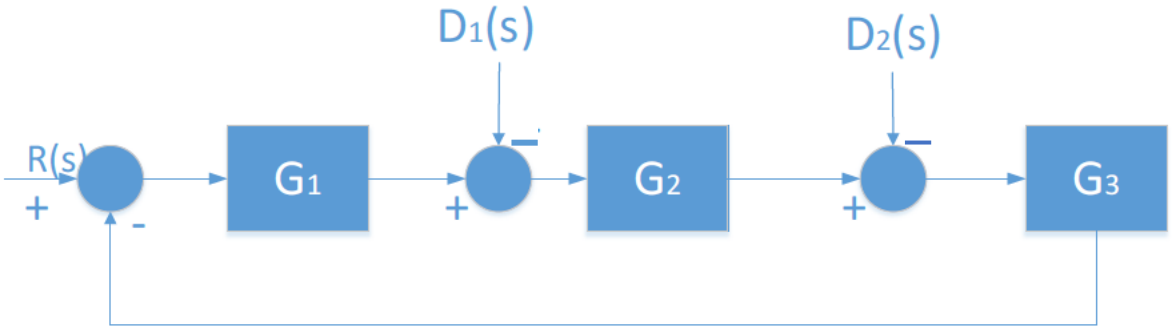
Where:

- $K_{p1} = \lim_{s\to 0} G_1(s)$
- $K_{p2} = \lim_{s\to 0} G_2(s)$
- $K_{v1} = \lim_{s\to 0} sG_1(s)$
- $K_{a1} = \lim_{s\to 0} s^2G_1(s)$
- $G(s) = G_1(s)G_2(s)$

Observations:

1) The determinant factor is the type of transfer function on the left of the disturbance (G_1 in this case). For example, G_2 could be type 2 but if G_1 has no integrator, the total steady state error cannot be zero if there is a step disturbance.
2) If we want zero error due to the disturbance, we need at least as many integrators in $G_1(s)$ as there are in the disturbance.
3) If $G_1(s)$ has two or more integrators less than the integrators in D(s), then the error will be infinite.
4) Remember that this table is only valid if the system is stable because the final value theorem requires all poles to be in the LHP.

Problem 1:
In the following system assume the systems is stable,



Where

$$R(s) = \frac{r_0}{s}$$
$$D_1(s) = \frac{d_1}{s}$$
$$D_2(s) = \frac{d_2}{s}$$

1) And G_1, G_2 and G_3 are

$$G_1(s) = \frac{3}{s+4}$$
$$G_2(s) = \frac{3(s+2)}{s^2+5s+10}$$
$$G_3(s) = \frac{s+10}{s+2}$$

Find the steady state errors due to R(s), $D_1(s)$ and $D_2(s)$ and the total error assuming the closed system is stable.

2) Now assume that

$$G_1(s) = \frac{3}{s+4}$$
$$G_2(s) = \frac{3(s+2)}{s^2+5s+10}$$
$$G_3(s) = \frac{s+10}{s^2+2s}$$

Find the steady state errors due to R(s), $D_1(s)$ and $D_2(s)$ and the total error assuming the closed system is stable.

3) Now assume

$$G_1(s) = \frac{3}{s+4}$$
$$G_2(s) = \frac{3(s+2)}{s^2+5s}$$
$$G_3(s) = \frac{s+10}{s+2}$$

Find the steady state errors due to R(s), $D_1(s)$ and $D_2(s)$ and the total error assuming the closed system is stable.

4) Now assume that

$$G_1(s) = \frac{3(s+1)}{s}$$
$$G_2(s) = \frac{3(s+2)}{s^2+5s+10}$$
$$G_3(s) = \frac{s+10}{s+2}$$

Find the steady state errors due to R(s), $D_1(s)$ and $D_2(s)$ and the total error assuming the closed system is stable.